Variational Disparity and Optical Flow Estimation

In that Order

James Lyndon Gray, 2025-03-11

Agenda

- History of Variational Methods in Optical Flow and Disparity estimation
- Variational Methods
- Variational Disparity/Depth Estimation
  - Problem Formulation
  - Posing Disparity Estimation as an optimisation problem
  - How to discretise the optimisation problem
- Optical Flow
  - Wow this is actually the same, but not quite?

History of Variational Methods

in Disparity and Optical Flow Estimation

Determining Optical Flow

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ABSTRACT
Optical flow cannot be computed locally, since only one independent measurement is available from the image sequence at a point, while the flow velocity has two components. A second constraint is needed. A method for finding the optical flow pattern is presented which assumes that the apparent velocity of the brightness pattern varies smoothly almost everywhere in the image. An iterative implementation is shown which successfully computes the optical flow for a number of synthetic image sequences. The algorithm is robust in that it can handle image sequences that are quantized rather coarsely in space and time. It is also insensitive to quantization of brightness levels and additive noise. Examples are included where the assumption of smoothness is violated at singular points or along lines in the image.

This is really the first paper and it laid the foundation for research into Optical Flow. It's from 1981!

High Accuracy Optical Flow Estimation Based on a Theory for Warping ⋆

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Abstract. We study an energy functional for computing optical flow that combines three assumptions: a brightness constancy assumption, a gradient constancy assumption, and a discontinuity-preserving spatio-temporal smoothness constraint. In order to allow for large displacements, linearisations in the two data terms are strictly avoided. We present a consistent numerical scheme based on two nested fixed point iterations. By proving that this scheme implements a coarse-to-fine warping strategy, we give a theoretical foundation for warping which has been used on a mainly experimental basis so far. Our evaluation demonstrates that the novel method gives significantly smaller angular errors than previous techniques for optical flow estimation. We show that it is fairly insensitive to parameter variations, and we demonstrate its excellent robustness under noise.

This paper from 2004 improved things considerably and nearly every variational paper since then references it. They use robust statistics and a new linearisation approach.

History of Variational Methods
in Disparity and Optical Flow Estimation

Optic Flow Goes Stereo: A Variational Method for Estimating Discontinuity-Preserving Dense Disparity Maps

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Abstract. We present a novel variational method for estimating dense disparity maps from stereo images. It integrates the epipolar constraint into the currently most accurate optic flow method (Brox *et al.* 2004). In this way, a new approach is obtained that offers several advantages compared to existing variational methods: (i) It preservers discontinuities very well due to the use of the total variation as solution-driven regulariser. (ii) It performs favourably under noise since it uses a robust function to penalise deviations from the data constraints. (iii) Its minimisation via a coarse-to-fine strategy can be theoretically justified. Experiments with both synthetic and real-world data show the excellent performance and the noise robustness of our approach.

This is an early paper (2005) where the variational optical flow approach gets adapted to stereo disparity estimation.

Multi-view Scene Flow Estimation: A View Centered Variational Approach

Tali Basha · Yael Moses · Nahum Kiryati

ELSEVIER

Physica D: Nonlinear Phenomena
Volume 60, Issues 1–4, 1 November 1992, Pages 259-268

Nonlinear total variation based noise removal algorithms ☆

Leonid I. Rudin [2], Stanley Osher, Emad Fatemi [2]

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https://doi.org/10.1016/0167-2789(92)90242-F

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Abstract

A constrained optimization type of numerical algorithm for removing noise from images is presented. The total variation of the image is minimized subject to constraints involving the statistics of the noise. The constraints are imposed using Lanrange multipliers. The solution is obtained using the gradient-projection method. This amounts to solving a time dependent partial differential equation on a manifold determined by the constraints. As $t \to \infty$ the solution converges to a steady state which is the denoised image. The numerical algorithm is simple and relatively fast. The results appear to be state-of-the-art for very noisy images. The method is noninvasive, yielding sharp edges in the image. The technique could be interpreted as a first step of moving each level set of the image normal to itself with velocity equal to the curvature of the level set divided by the magnitude of the gradient of the image, and a second step which projects the image back onto the constraint set.

You can also use variational approaches to do scene flow and all kinds of similar problems!

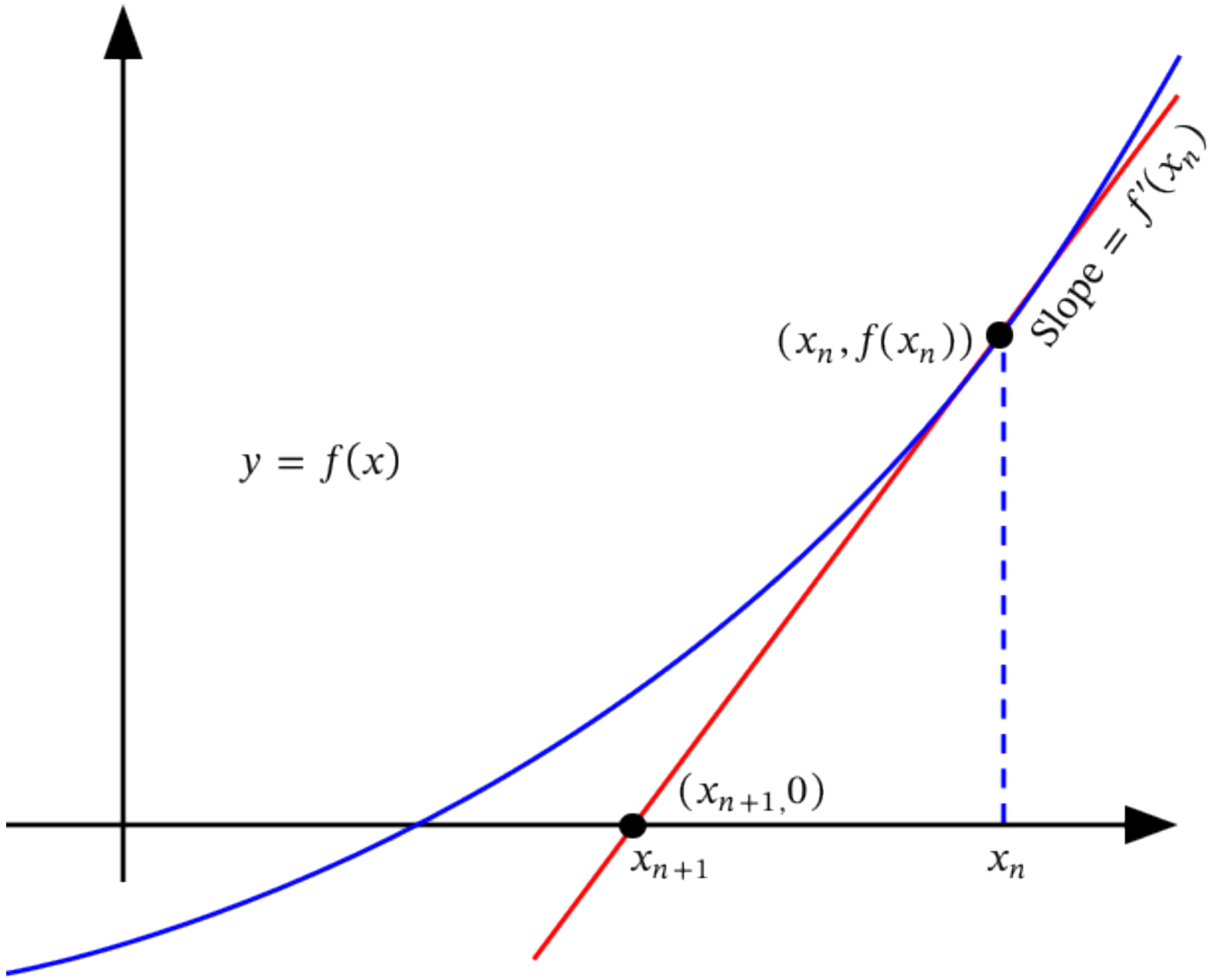
Variational Methods

- Optimisation problems seek to minimise a cost/energy function.
- One way to do this is to consider multiple candidates for the global minimum. You would then select the best candidate.
- Variational approaches are different. Instead of considering multiple candidates, you solve the Euler-Lagrange equations and arrive at the solution.
- Solving the Euler-Lagrange equation is often hard.
  - So, linearise them successively.
  - Consider a single candidate and then move to where the linearised Euler-Lagrange equations tell you, and you repeat the process until you get the right answer.

Approx. Variational Methods

Similar to Newton's Method

- The process of successive approximation is conceptually very similar to Newton's Method.
- You can probably pose solving the Euler-Lagrange equations as something that could be solved by Newton's method.



$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

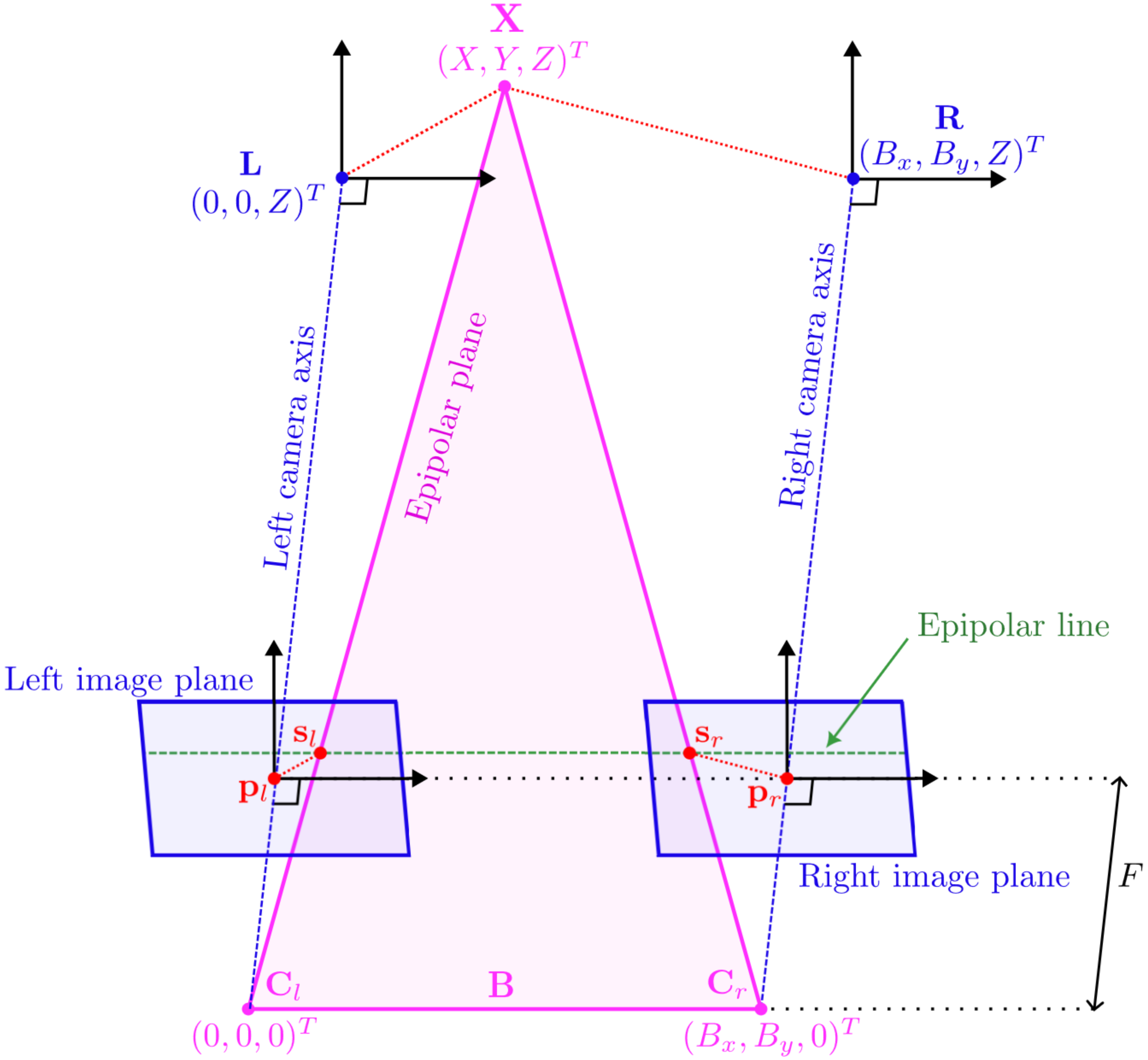
Pros & Cons of Variational Methods

Why even apply them to Disparity and Optical Flow estimation?

- Variational Methods only consider one candidate at a time, so they are cheap in terms of memory, compared to keeping a cost volume in memory and searching that (often used in ML schemes in optical flow/disparity estimation).
- Very few parameters, sometimes just one.
- Possible to debug.
- However, because they rely on successive linearisations, they might be limited by the accuracy of these linearisations.
- You have to be able to express cues in terms of a sensible energy cost. Figuring this out can be tricky.

Disparity/Depth Estimation

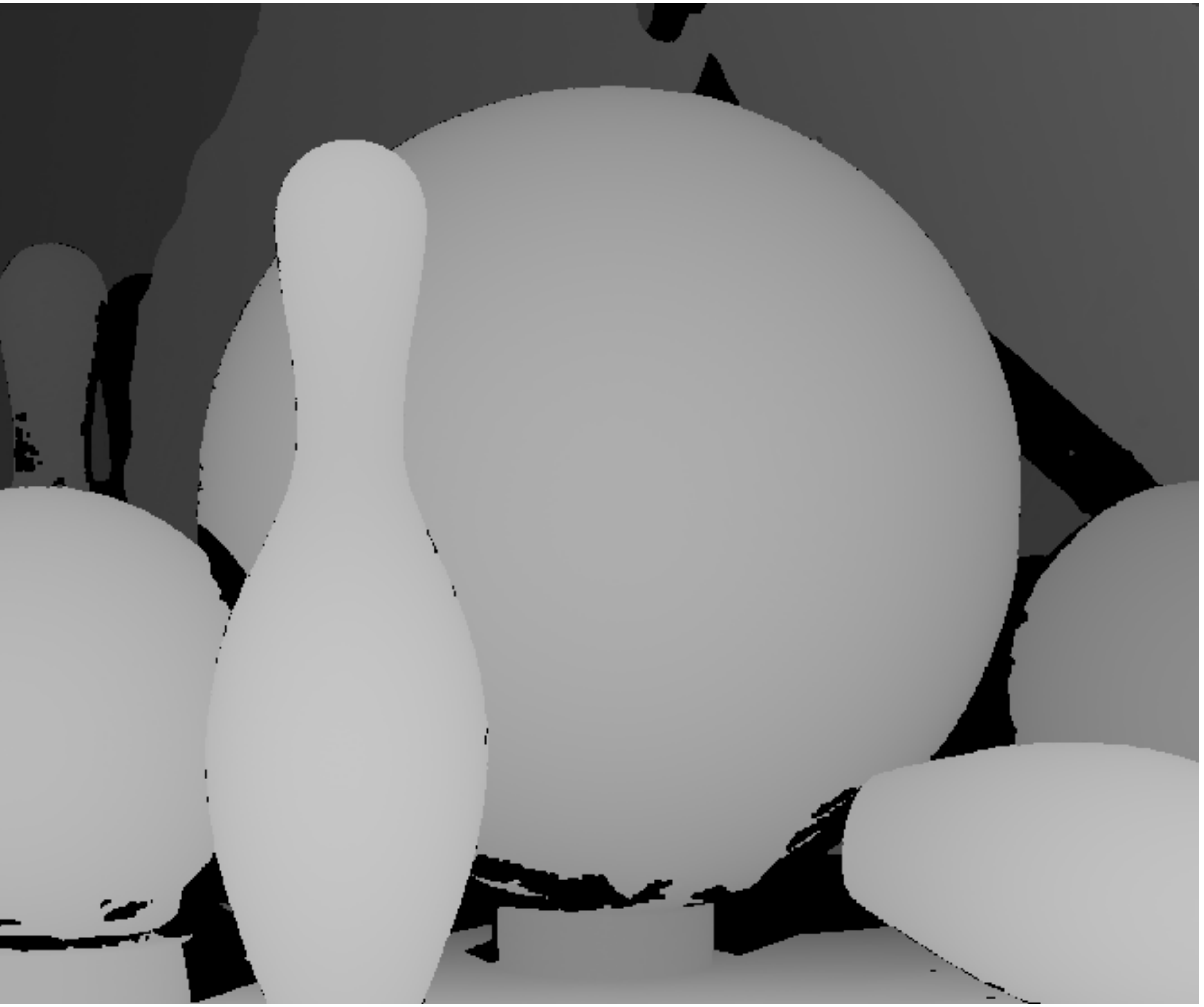
- Let's start with Stereo.
- Our goal is to estimate z coordinate of $\mathbf{X}$, Z, and do it for *every* point in the "reference" image (usually left).
- We can use similar triangles to arrive at the equation, $\mathbf{v} = \mathbf{B}F/Z$, where $\mathbf{v}$ is the disparity vector $(\mathbf{s}_l - \mathbf{s}_r)$, F is the focal length and $\mathbf{B}$ is the baseline vector.
- So if we know which points in each image correspond to X, finding Z is easy.
- The hard bit is finding the correspondences.



Some Sensible Assumptions

- So to establish correspondences, we need to make some basic assumptions:
  - Brightness constancy. Corresponding points should look the same in both views.
  - The disparity field is smooth.
- We need to write these assumptions in terms of an energy minimisation problem.
  - Brightness constancy constraint: $E_d = \left(I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s})) - I_l(\mathbf{s})\right)^2$
  - Smoothness: $E_r = |\nabla r(\mathbf{s})|^2$. N.B. r is the reciprocal depth or normalised disparity. Its relationship with the disparity vector is given by, $\mathbf{v}(\mathbf{s}) = \mathbf{B}r(\mathbf{s})$.
  - It's conceivable that you could try to solve $I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s})) = I_l(\mathbf{s})$ for $r(\mathbf{s})$ pointwise without even making the smoothness assumption. This is likely to be noisy however and you'd get poor results in areas with low texture.
  - How sensible are these assumptions?

Mostly?



Overall Energy Function

- We can now write an overall energy function:

$$E_{full} = \int_{\Omega} \left(I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s})) - I_l(\mathbf{s})\right)^2 + \alpha \left(\left(\frac{\partial r(\mathbf{s})}{\partial s_1} \right)^2 + \left(\frac{\partial r(\mathbf{s})}{\partial s_2} \right)^2 \right) d\mathbf{s},$$

  where α is a parameter that determines the strength of the smoothness assumption and Ω is the image domain.

- Remember, our goal is to minimise E_{full}.

A Brief Note on The Euler-Lagrange Equation

Brought to you by Wikipedia

$$E_{full} = \int_{\Omega} \left(I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s})) - I_l(\mathbf{s})\right)^2 + \alpha \left(\left(\frac{\partial r(\mathbf{s})}{\partial s_1} \right)^2 + \left(\frac{\partial r(\mathbf{s})}{\partial s_2} \right)^2 \right) ds$$

- I keep referring back to the wikipedia page on this, because it's quite good.
- The way to apply this is $r(s_1, s_2) \rightarrow f$ and $s_j \rightarrow x_j$.
- You can try this yourself.

Single function of several variables with single derivative

[edit source]

A multi-dimensional generalization comes from considering a function on n variables. If Ω is some surface, then

$$I[f] = \int_{\Omega} \mathcal{L}(x_1, \ldots, x_n, f, f_1, \ldots, f_n)\, \mathrm{d}\mathbf{x}\,; \quad f_j := \frac{\partial f}{\partial x_j}$$

is extremized only if *f* satisfies the partial differential equation

$$\frac{\partial \mathcal{L}}{\partial f} - \sum_{j=1}^{n} \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial f_j} \right) = 0.$$

When $n = 2$ and functional $\mathcal{I}$ is the energy functional, this leads to the soap-film minimal surface problem.

The Euler-Lagrange Equation

What a mess

$$E_{full} = \int_\Omega \left(I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s})) - I_l(\mathbf{s})\right)^2 + \alpha \left(\left(\frac{\partial r(\mathbf{s})}{\partial s_1}\right)^2 + \left(\frac{\partial r(\mathbf{s})}{\partial s_2}\right)^2 \right) d\mathbf{s}$$

The Euler-Lagrange equation for this is:

$$(I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s})) - I_l(\mathbf{s}))\frac{\partial I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s}))}{\partial r} - \alpha \nabla \cdot \nabla r(\mathbf{s}) = 0$$

This is quite non-linear and if you don't have a nice form for $\frac{\partial I_r(\mathbf{s} + \mathbf{B}r(\mathbf{s}))}{\partial r}$, solving this will be a pain.

Linearise the Energy Function?

1st Order Taylor Series Expansion

$$E_{full} = \int_{\Omega} \left(\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle r(\mathbf{s}) + I_r(\mathbf{s}) - I_l(\mathbf{s}) \right)^2 + \alpha \left(\left(\frac{\partial r(\mathbf{s})}{\partial s_1} \right)^2 + \left(\frac{\partial r(\mathbf{s})}{\partial s_2} \right)^2 \right) d\mathbf{s},$$

where $\langle , \rangle$ is the inner or dot product.

This should make things a bit easier, let's try again:

$$\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle \left(\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle r(\mathbf{s}) + I_r(\mathbf{s}) - I_l(\mathbf{s}) \right) - \alpha \nabla \cdot \nabla r(\mathbf{s}) = 0$$

If you discretise the $\nabla \cdot \nabla$ operator, this becomes a set of coupled linear equations that you can solve.

But how?

- We can discretise $-\nabla \cdot \nabla r(\mathbf{s})$ by treating the $-\nabla \cdot \nabla$ as a Finite Impulse Response filter, performing convolution. Essentially, weighted sum.
- There are two common choices for the operator:

$$\begin{pmatrix} & -1 & \\ -1 & 4 & -1 \\ & -1 & \end{pmatrix}, \quad \begin{pmatrix} -1 & -2 & -1 \\ -2 & 12 & -2 \\ -1 & -2 & -1 \end{pmatrix}$$

- For simplicity, we will use the first one.

Discretising the Euler-Lagrange Equations

- We have, $\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle \big(\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle r(\mathbf{s}) + I_r(\mathbf{s}) - I_l(\mathbf{s}) \big) - \alpha \nabla \cdot \nabla r(\mathbf{s}) = 0$, but can we write it as a matrix equation?
- Let's do some rearranging: $\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle^2 r(\mathbf{s}) - \alpha \nabla \cdot \nabla r(\mathbf{s}) = \langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle (I_l(\mathbf{s}) - I_r(\mathbf{s}))$.
- This can be written as $A\mathbf{r} = \mathbf{b}$, where $\mathbf{r}$ is a vector of length $n =$ height $\times$ width, so $\mathbf{b}$ is also of length n and A is $n \times n$ in size.
- $\mathbf{r}$ is simply the $r(\mathbf{s})$ values for each pixel in order.
- $\mathbf{b}$ is $\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle (I_l(\mathbf{s}) - I_r(\mathbf{s}))$ for each pixel in order.
- The A matrix is a bit more complex…

Discretising the Euler-Lagrange Equations

- We need to write $\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle^2 r(\mathbf{s}) - \alpha \nabla \cdot \nabla r(\mathbf{s}) = \langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle (I_l(\mathbf{s}) - I_r(\mathbf{s}))$ as an $A\mathbf{r} = \mathbf{b}$ equation.

- Let's split A by taking, $A = A_d + \alpha A_s$. A_d is a pointwise term, so it's a diagonal matrix. i.e.

$$A_d = \begin{bmatrix} \langle \nabla I_r(0,0), \mathbf{B} \rangle^2 & & & \\ & \langle \nabla I_r(1,0), \mathbf{B} \rangle^2 & & \\ & & \ddots & \\ & & & \langle \nabla I_r(w,h), \mathbf{B} \rangle^2 \end{bmatrix}$$

- h, w is height and width.

What about A_s?

- Remember, we are trying to express a convolution operator as a (block-circulant) matrix. $A_s = \begin{bmatrix} C & -I & & & \\ -I & C & -I & & \\ & \ddots & \ddots & \ddots & \\ & & -I & C & -I \\ & & & -I & C \end{bmatrix}$, where $C = \begin{bmatrix} 4 & -1 & & & \\ -1 & 4 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 4 & -1 \\ & & & -1 & 4 \end{bmatrix}$.
- Each block is width $\times$ width in size.
- Note that we're ignoring edge conditions for simplicity.
- Now we know what all the bits of $A\mathbf{r} = \mathbf{b}$ are, we can use a matrix solver to solve it!
- NB: This matrix is sparse, symmetric and positive definite, which can inform which technique to use.

Summary So Far

- We began with a energy function E_{full}, that we linearised, i.e.

$$E_{full} = \int_{\Omega} \big(\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle r(\mathbf{s}) + I_r(\mathbf{s}) - I_l(\mathbf{s})\big)^2 + \alpha \left(\left(\frac{\partial r(\mathbf{s})}{\partial s_1} \right)^2 + \left(\frac{\partial r(\mathbf{s})}{\partial s_2} \right)^2 \right) d\mathbf{s},$$

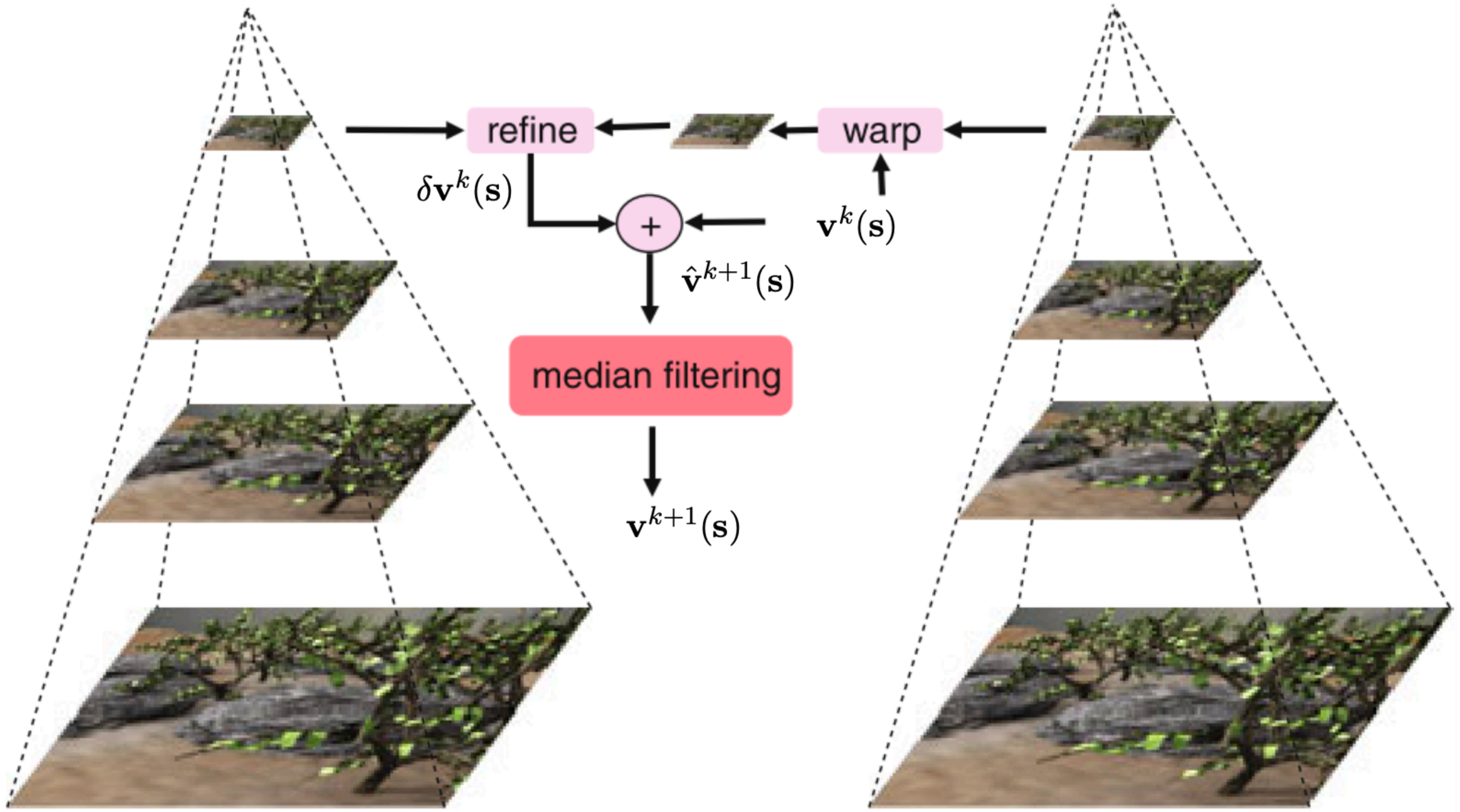
- We wrote down the Euler-Lagrange Equations that minimise E_{full}, i.e.

$$\langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle^2 r(\mathbf{s}) - \alpha \nabla \cdot \nabla r(\mathbf{s}) = \langle \nabla I_r(\mathbf{s}), \mathbf{B} \rangle (I_l(\mathbf{s}) - I_r(\mathbf{s}))$$

- We formulated the Euler-Lagrange Equations as a matrix equation of the form $A\mathbf{r} = \mathbf{b}$.

A Few Implementation Details

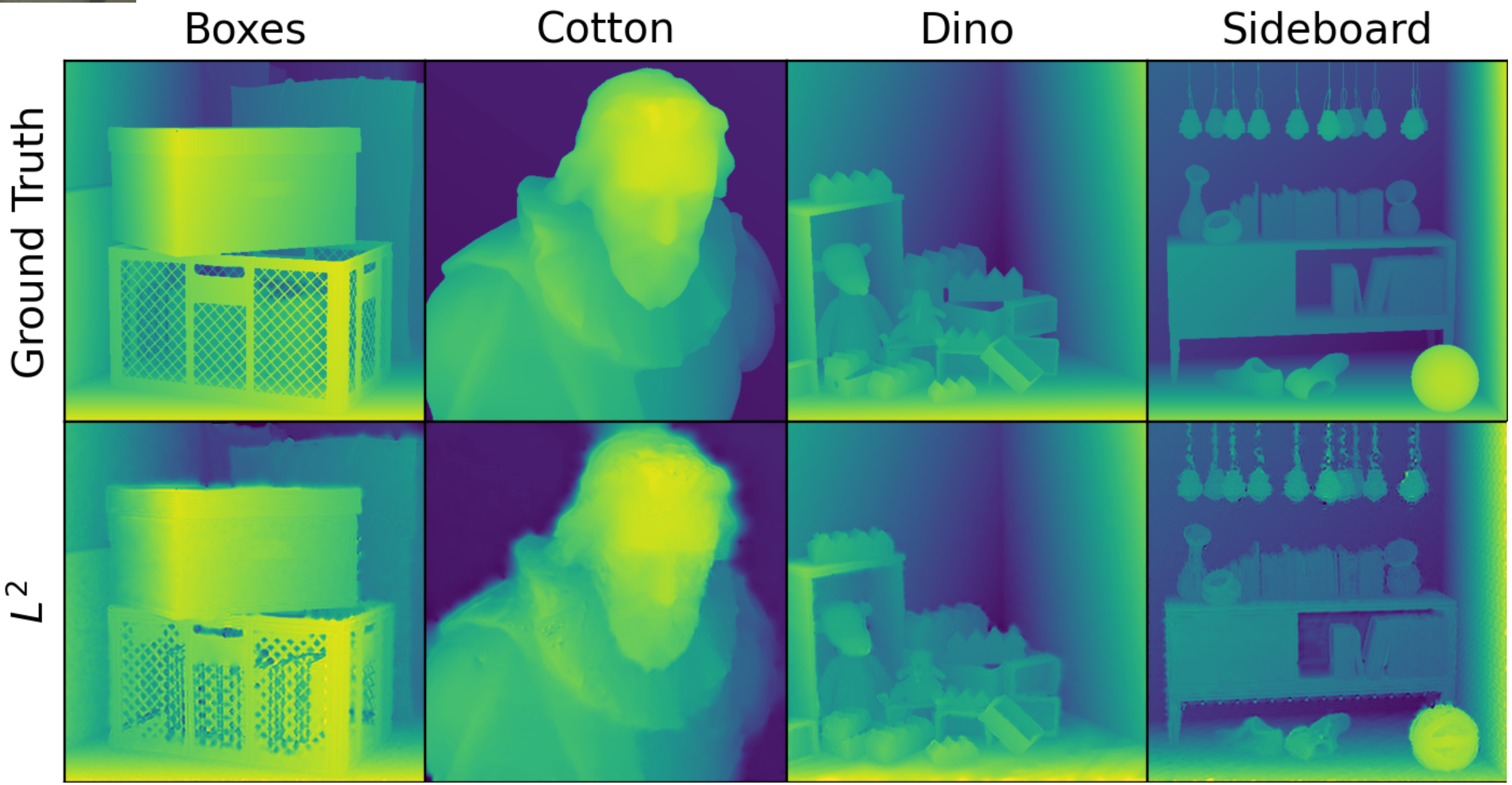
- I mentioned that this was an iterative procedure. How does this work?
- After each time we solve the Euler-Lagrange equations we have to warp the target image.
- Often we start at coarse resolutions and move to finer resolutions.
- Median filtering between iterations is useful.
- Instead of using L_2 loss functions, we can use robust statistics.



Although this diagram depicts the CTF approach for optical flow, the concept is the same for disparity estimation.

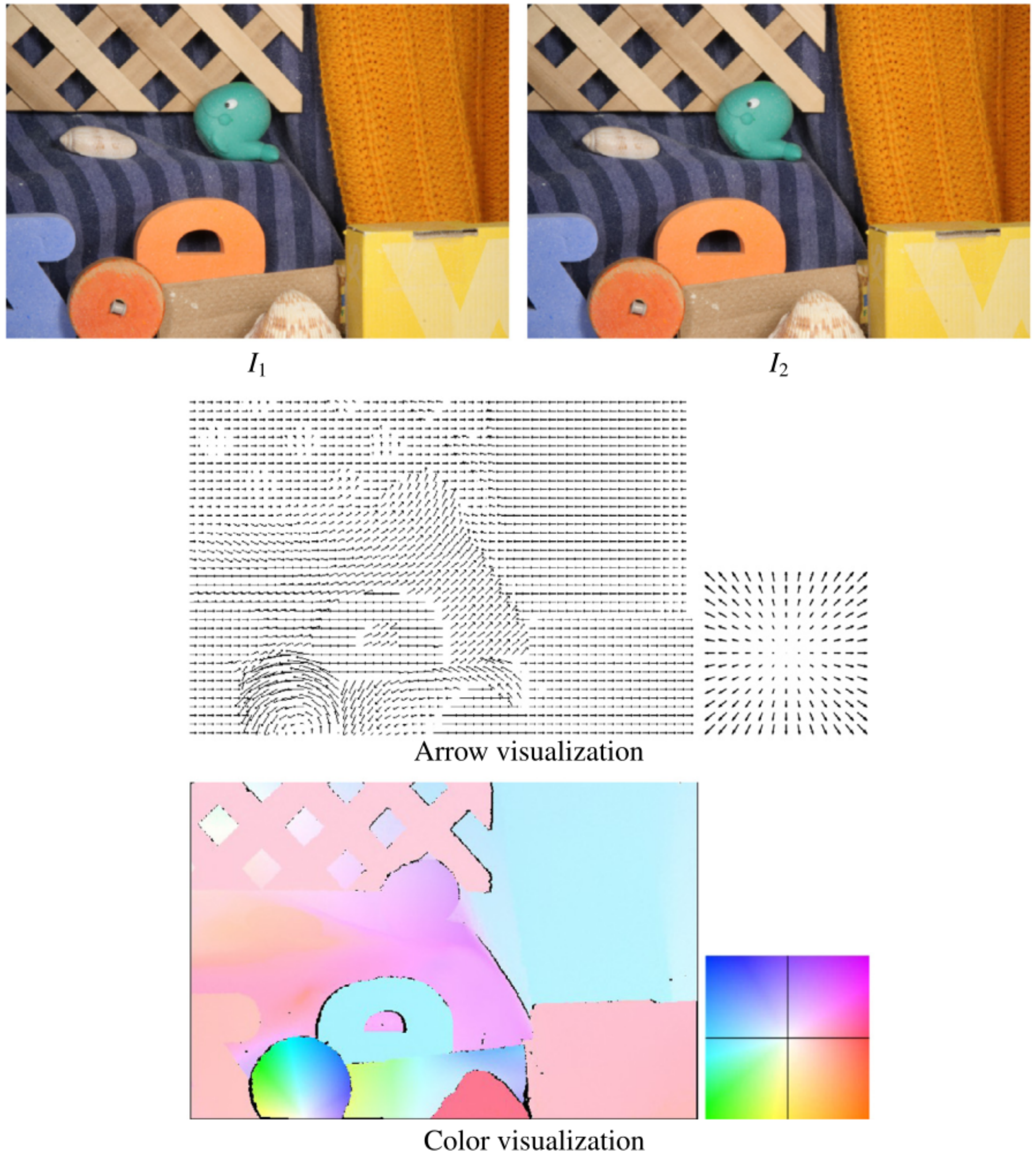
Does this work?

Yeah, it's okay?



What about Optical Flow?

- This is where you track the 2D projection of 3D scene motion onto the image plane. Where does each pixel move?
- Instead of one unknown per pixel, $r(\mathbf{s})$, we two unknowns, $\mathbf{v}(\mathbf{s}) = (v_1(\mathbf{s}), v_2(\mathbf{s}))$.
- $v_1(\mathbf{s})$ is the horizontal component and $v_2(\mathbf{s})$ is the vertical component of $\mathbf{v}(\mathbf{s})$.
- Unlike with disparity where it's conceivable that all you might need is the brightness constancy assumption, you need additional information.



The Aperture Problem

- Let's write down the brightness constancy constraint for optical flow: $E_d = (I(\mathbf{s} + \mathbf{v}(\mathbf{s}), t + 1) - I(\mathbf{s}, t))^2$.
- Linearising it give us:
  $E_d = \left(\langle \mathbf{v}(\mathbf{s}), \nabla I(\mathbf{s}, t)\rangle + I(\mathbf{s}, t + 1) - I(\mathbf{s}, t)\right)^2$
- That's one equation for two unknowns and this is what causes the Aperture Problem, which gives rise to the barber pole illusion.
- We need more information. Let's formulate the smoothness assumption for optical flow:
  $E_r = |\nabla v_1(\mathbf{s})|^2 + |\nabla v_2(\mathbf{s})|^2$.

Optical Flow Overall Energy Function

$$E_{full} = \int_{\Omega} \left(\langle \nabla I(\mathbf{s}, t), \mathbf{v}(\mathbf{s}) \rangle + I(\mathbf{s}, t+1) - I(\mathbf{s}, t) \right)^2 + \alpha \left(|\nabla v_1(\mathbf{s})|^2 + |\nabla v_2(\mathbf{s})|^2 \right) d\mathbf{s},$$

So what's the next step?
Refer to the Euler-Lagrange equation wikipedia page.

Several functions of several variables with single derivative [edit source]

If there are several unknown functions to be determined and several variables such that

$$I[f_1, f_2, \ldots, f_m] = \int_{\Omega} \mathcal{L}(x_1, \ldots, x_n, f_1, \ldots, f_m, f_{1,1}, \ldots, f_{1,n}, \ldots, f_{m,1}, \ldots, f_{m,n}) \,\mathrm{d}\mathbf{x}\,; \quad f_{i,j} := \frac{\partial f_i}{\partial x_j}$$

the system of Euler–Lagrange equations is[5]

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial f_1} - \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial f_{1,j}}\right) &= 0_1 \\
\frac{\partial \mathcal{L}}{\partial f_2} - \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial f_{2,j}}\right) &= 0_2 \\
\vdots \qquad \vdots \qquad & \vdots \\
\frac{\partial \mathcal{L}}{\partial f_m} - \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial f_{m,j}}\right) &= 0_m.
\end{aligned}$$

Optical Flow Euler-Lagrange Equations

- So if we use the formula on the previous slide, we get:

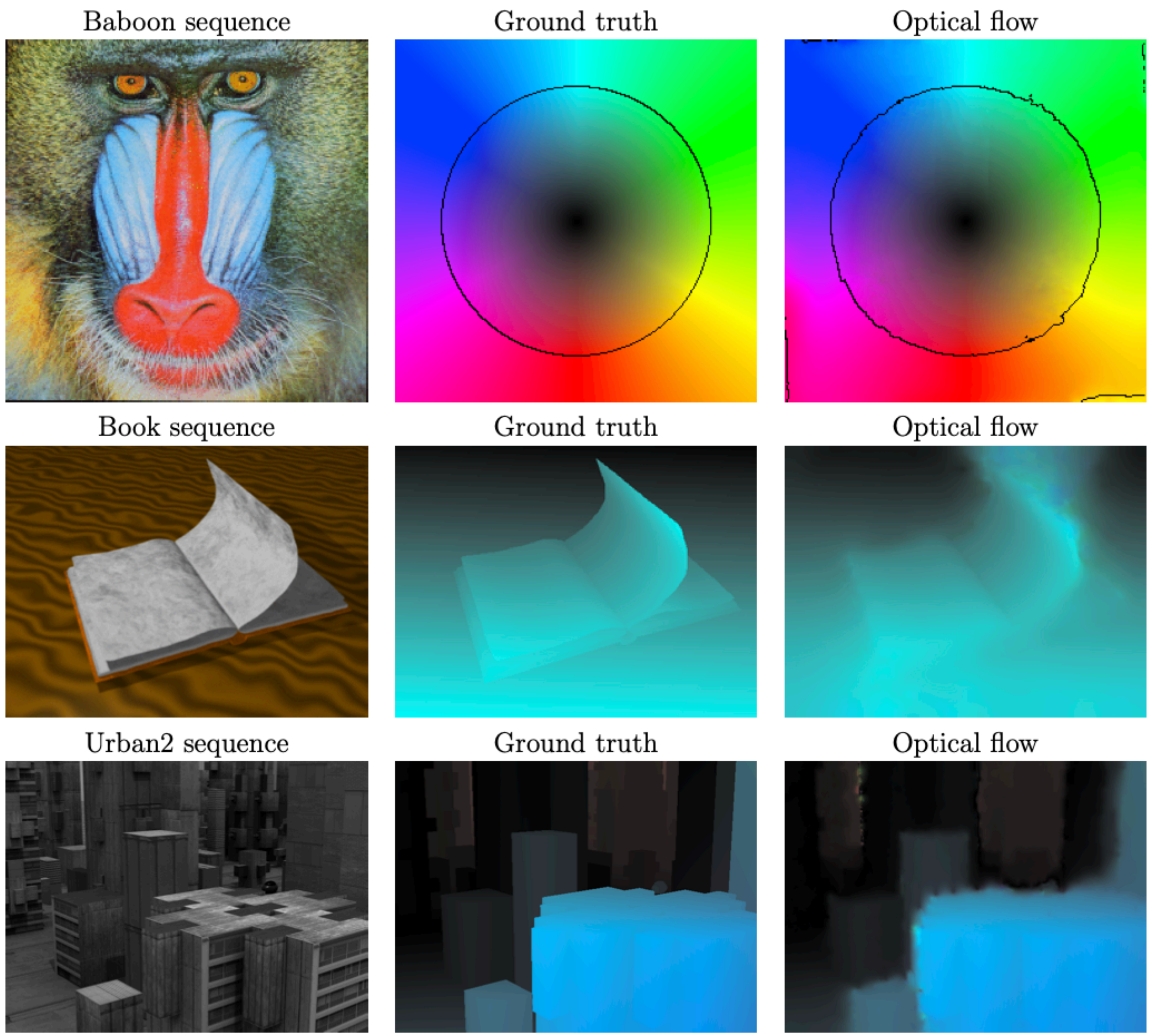
$$\frac{\partial I(\mathbf{s})}{\partial s_1}\Big(\langle \nabla I(\mathbf{s},t), \mathbf{v}(\mathbf{s})\rangle + I(\mathbf{s},t+1) - I(\mathbf{s},t)\Big) - \alpha\nabla \cdot \nabla v_1(\mathbf{s}) = 0,$$

$$\frac{\partial I(\mathbf{s})}{\partial s_2}\Big(\langle \nabla I(\mathbf{s},t), \mathbf{v}(\mathbf{s})\rangle + I(\mathbf{s},t+1) - I(\mathbf{s},t)\Big) - \alpha\nabla \cdot \nabla v_2(\mathbf{s}) = 0.$$

- These are a system of coupled linear equations, that we can write as a matrix equation of the form, $A\mathbf{x} = \mathbf{b}$.
- I'm not going to go through this whole process, because it's very similar to the disparity case.
- What you do is set, $\mathbf{x} = \begin{pmatrix}\mathbf{v_1}\\ \mathbf{v_2}\end{pmatrix}$, where $\mathbf{v_i}$ is just $v_i(\mathbf{s})$ for each pixel in order etc.

Does it work?

- Let's have a look: https://www.ipol.im/pub/art/2013/20/
- Sometimes the demo doesn't work, so here are some of the plots.
- Seems okay?



Where to Look for More Info

- The wikipedia page on Optical Flow is good for an overview.
- My thesis: https://doi.org/10.26190/unsworks/30875
  - Background chapters and sections might be useful
- Meinhardt-Llopis, E., & Sánchez, J. (2013). Horn-schunck optical flow with a multi-scale strategy. Image Processing on line.
  - This is a really good paper in terms of explaining how the whole process works. The website is also pretty neat, it lets you run some demos: https://www.ipol.im/pub/art/2013/20/
  - It's the only paper I've seen to include the ugly matrix work.
- Horn, B. K., & Schunck, B. G. (1981). Determining optical flow. Artificial intelligence, 17(1-3), 185-203.
  - The original paper on optical flow
- Tran, T. H., Wang, Z., & Simon, S. (2017, July). Variational disparity estimation framework for plenoptic images. In 2017 IEEE International Conference on Multimedia and Expo (ICME) (pp. 1189-1194). IEEE.
  - A solid paper on how to apply the variational method to disparity estimation with light fields.

Where to Look for More Info

- Books? I haven't read these in depth, but maybe they're useful?
  - G. Aubert and P. Kornprobst, Mathematical Problems in Image Processing: Partial Differential Equations and the Calculus of Variations (2nd edition).
    - This has a section on Optical Flow: 5.3.2. It also provides mathematical background on the Calculus of Variations in Chapter 2.
- R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed.
  - Section 9.3 discusses optical flow.
- More perhaps?